

Basil Aranda

Active Security Clearance | Huntington Beach, CA. 92647 | thymetravel.github.io

Work Experience

Anduril Industries

May 2025 – Present

Manufacturing Engineer, Sentry

- Designed and fabricated critical tooling fixtures, shop aids, and test equipment using CAD (SolidWorks, NX) and rapid prototyping (3D printing, CNC machining), preventing production line-down and enabling assembly of complex subassemblies
- Root caused field failures and drove ECNs/MCNs for multiple subsystems using hands-on diagnostics, engineering fundamentals (structures, thermal, electrical), and manufacturing knowledge
- Scaled production capacity 4x (10 to 40 units/month) for complex electromechanical systems through process optimization, manufacturing design improvements, and hands-on technician support
- Created custom payload spoofer fixtures for system validation testing, replacing \$300K+ deployable hardware while enabling full qualification campaigns for flight hardware

Lockheed Martin Skunk Works

August 2023 – April 2025

Manufacturing Engineer, ADP

- Responsible manufacturing engineer for Advanced Development Program (ADP) in charge of retrofit and manufacturing design of light unmanned jet aircraft
- Create corrective and preventative action plans for recurring high-cost and mission critical product defects saving hundreds of thousands of dollars while ensuring adherence to AS9100
- Support the maintenance, repair, and overhaul (MRO) of specialized legacy aircraft, vital to the Pentagon's security portfolio
- Drive and own multiple million-dollar capital projects for composite production including research, use, and improvement

Aerojet Rocketdyne

May 2022 – August 2022

Project & Component Engineering Intern

- Improved process efficiency of Manufacturing Bill of Materials creation by over 99.6% utilizing advanced data science
- Created Qualification and Acceptance standards utilizing strict SMC and AIAA requirements for testing and flight readiness

NASA Armstrong Flight Research Center

January 2022 – May 2022

Controls System Engineering Intern

- Developed electromechanical interface and flight control system for human-sized quadcopters
- Heavy utilization of MATLAB and Simulink to provide HITL validation and testing from controls engineering first-principles
- Modeled and simulated flight controllers and high power motors using control systems and actual hardware frequency
- Worked with and designed PWM-based motors with transient disturbances and damping

Education

California State Polytechnic University, Pomona

May 2023

B.S. Aerospace Engineering, magna cum laude

GPA: 3.76

Astronomy Minor

Skills & Abilities

Manufacturing: Sheet metal, composites, 3D printing, injection molding, welding, soldering | Machine shop proficiency

Software: NX, Python, C++, MATLAB, C, SolidWorks, CREO, Fusion 360, Altium, Simulink, Adobe Premiere

Industrial: Project Development, Systems Engineering, Technical Writing, Manufacturing Design, Controls Engineering

Publications

Aranda, Basil P., et al. "Design of Dual Mars Ascent Vehicle MAVERICK." *American Institute of Aeronautics and Astronautics* AIAA 2023-4779, 2023, <https://doi.org/10.2514/6.2023-4779>.